home affairs (MHA). After the revelation, the company's tender for supplying CCTVs to the J&K Lieutenant Governor's office in Srinagar was cancelled. However, no further action was taken.

THE NEED FOR RAPID REMEDY

In the absence of any specific guidelines for the home ministry, many experts are now calling for additional terms and conditions to be implemented for the entire procurement process of VSS systems to ensure foolproof safety from espionage. National Cyber Security Coordinator Pant says he has funded the government-owned Centre for Development of Telematics (C-DOT) to work out a national trust centre for software and hardware testing to verify devices being purchased by sensitive government departments. The other example he cites is Singapore, which has come out with IoT (Internet of Things) star ratings that indicate whether or not a device has been cleared for snooping bugs.

Experts point to official complacency about the compulsory compliance of CCTVs under BIS. Existing standards for CCTV systems are the same that apply to mobile phones, cash registers, laptops, set-top boxes, power banks and the like, whereas the need of the hour, they say, are regulations that specifically address national security concerns. "Nowhere does it talk about its national security aspects. BIS only mention human hazards like fire or their durability," says N.K. Goel, chairman of the Telecom Equipment Manufacturers Association of India (TEMA).

EXISTING STANDARDS FOR CCTVS ARE THE SAME AS FOR CELLPHONES, LAPTOPS ETC., WHEREAS THE NEED OF THE HOUR, EXPERTS SAY, ARE REGULATIONS THAT ADDRESS NATIONAL SECURITY CONCERNS

While flagging national security concerns with regard to surveillance systems, military experts also point out how various household devices are increasingly being connected to the internet—from televisions to washing machines—and are thus susceptible to deployment for data theft. "Xiaomi, a top Chinese manufacturer of phones and other gadgets, sells a fitness band with a military-grade sensor and 30-day battery life for $35. These watches often track movements, heart rates and sleep patterns. So, there is always a fear of being snooped," says a military observer.

Lt Gen. Bhatia says that the Chinese can easily amass crucial information through data fusion by combining inputs from CCTVs with other available data (from mobile phones, for instance). "China can easily have remote control access to these surveillance devices," he says. "The threat is imminent; besides making changes in our procurement process, we need to make our troops aware of new ways of snooping by China."

In addition to the national security threat, the market dominance of Chinese-made CCTVs is a lost opportunity for India to become a net exporter of surveillance hardware through manufacturing, rather than just the assembly of Chinese goods. According to a report by Market Research Future, a Pune-based market analysis company, the global CCTV market is expected to grow to approximately $46.52 billion (Rs 3.8 lakh crore) at a 13.1 per cent CAGR (compound annual growth rate) between 2020 and 2030. Industry insiders claim that due to Chinese imports, India has almost no share in the above. However, things could change if cheap Chinese imports flooding our market are brought to a halt and Indian companies are encouraged.

Now that alarm bells have started ringing in the security establishment, a department in North Block is looking at curbing potential Chinese spying through surveillance devices, but the widespread network of suspicious CCTVs has made it an uphill task. Intelligence agencies have confirmed instances where they found that devices have chips installed in them that are capable of sending back information to China. However, if the Centre puts an immediate ban on everything related to the Chinese telecom network, it would affect its own surveillance mechanism. Replacing entire surveillance systems will also mean incurring an additional financial burden. Experts, therefore, recommend taking action in a phased manner, beginning with highly sensitive security establishments and then gradually phasing it out in other government institutions.

"What we are trying to tweak are the Chinese chips embedded in cameras. In some instances, it was found these chips are reprogrammable by Indian software. If that is done, the Chinese snooping menace can be curtailed," says a cyberexpert working with the government, speaking on condition of anonymity. "Banning (of Chinese-made CCTVs) is a political step. We need to make sure there is no backdoor entry of Chinese intelligence into surveillance devices," he says, adding that the absence of any clear-cut data protection law further complicates the situation. The more pressing concern at the moment, though, is the effective weeding out of Chinese surveillance systems that can compromise the nation's security. As the clock ticks on, an effective and foolproof legal response with adequate action on the ground is awaited. ■